

Data Cleaning & Preparation — Change Log

Project 1 · DecodeLabs Data Analytics Internship (Batch 2026)
Dataset: raw_orders.xlsx → cleaned_orders.xlsx

Summary

Metric	Value
Raw rows	238
Cleaned rows	220
Rows removed (duplicates)	18
Duplicate Order_ID error rate	0.0%
Incorrect date format rate	0.0%
Remaining missing values	0

Verification Gate for Project 2 passed: 0% duplicate unique identifiers and 0% incorrectly formatted dates confirmed above.

Change Log

Change ID	Description	Impact	Status
CR001	Imputed 'Qty' using Median (2.0)	Preserved 9 records	Resolved
CR002	Imputed 'Value' using per-product Median	Preserved 7 records	Resolved
CR003	Imputed 'City' using Mode ('Islamabad')	Preserved 12 records	Resolved
CR004	Imputed 'Status' using Mode ('Delivered')	Preserved 7 records	Resolved
CR005	Standardized 'City' to one canonical spelling per city	Corrected 177 records	Resolved
CR006	Standardized 'Status' casing & whitespace	Corrected 121 records	Resolved
CR007	Converted 'Order_Date' to ISO 8601 (YYYY-MM-DD)	Corrected 192 records	Resolved
CR008	Enforced numeric precision (Value: 2 decimals, Qty: integer)	Applied to all 238 records	Resolved
CR009	Removed exact duplicate rows	Removed 18 records	Resolved
CR010	Removed remaining duplicate Order_IDs (kept first record)	Removed 0 records	Resolved

Methodology Notes

- Missing values: numeric fields (Qty, Value) imputed with median; categorical fields (City, Status) imputed with mode — rows preserved rather than deleted, per the 'Strategic Imputation' guideline (listwise deletion reduces statistical power).
- Duplicates: removed in two passes — exact full-row duplicates first, then any remaining rows sharing an Order_ID (keeping the first occurrence) so Order_ID is a true unique key.

- Formatting: City names collapsed to one canonical spelling per city (case, abbreviation, and whitespace variants merged); Status trimmed and title-cased; Order_Date converted to ISO 8601 (YYYY-MM-DD) from five mixed source formats; Value rounded to 2 decimals; Qty cast to integer.

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